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1 Evolution of the Baryonic Tully-Fisher Relation with Redshift

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March 2026

1.1 Abstract

The Baryonic Tully-Fisher Relation (BTFR) — $M_{bar} \propto v^4$ — is one of the tightest scaling relations in extragalactic astronomy. We derive its evolution with redshift from the assumption that the MOND acceleration scale tracks cosmic density: $a_0(z) = a_0(0) \times E(z)$. This predicts that at fixed rotation velocity, high-redshift galaxies have **less** baryonic mass than local galaxies:

At z = 2, this corresponds to a **−0.48 dex** offset (factor of 3×). We test this against kinematic data from KMOS3D, SINS, and ALPINE surveys and find:

- **Evolving a_0: $\chi^2 = 119$**
- **Constant a_0: $\chi^2 = 417$**
- **Improvement: 3.5×**

This is a falsifiable prediction: if high-redshift BTFR shows no offset from local, or the wrong sign, the framework is ruled out.

1.2 1. Introduction

1.2.1 1.1 The Baryonic Tully-Fisher Relation

The BTFR relates baryonic mass to rotation velocity:

where $a_0 \approx 1.2\times10^{-10}$ m/s² is the MOND acceleration scale.

Key properties (local universe): - Slope: 4.0 ± 0.1 (McGaugh et al. 2000) - Scatter: 0.1 dex (Lelli et al. 2016) - Applies to all rotationally-supported galaxies

1.2.2 1.2 The Question

Does the BTFR evolve with redshift?

Model	Prediction
ΛCDM	Complex evolution (halo-dependent)
Standard MOND	No evolution (a_0 = constant)
Evolving a_0	Systematic offset with z

1.2.3 1.3 Our Prediction

If $a_0(z) = a_0(0) \times E(z)$, then at fixed velocity:

Therefore:

1.3 2. Quantitative Predictions

1.3.1 2.1 The Evolution Function

$E(z) = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$ with $\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$

Redshift	E(z)	Δlog M_bar (dex)	Mass Ratio
z = 0	1.00	0.00	1.0×
z = 0.5	1.32	–0.12	0.76×
z = 1	1.79	–0.25	0.56×
z = 2	3.03	–0.48	0.33×
z = 3	4.57	–0.66	0.22×
z = 5	8.29	–0.92	0.12×

1.3.2 2.2 Physical Interpretation

At fixed rotation velocity v: - **Higher $a_0(z)$** → stronger MOND effect - **Stronger MOND** → less baryonic mass needed to produce same v - **Result:** High-z galaxies appear "under-massive" compared to local BTFR

This is **counterintuitive** — naively one might expect high-z galaxies to be *more* massive (more gas, younger). But the BTFR offset goes the *opposite* direction.

1.3.3 2.3 Not a Free Fit

The evolution E(z) is calculated from: - $\Omega_m = 0.315$ (Planck 2018) - $\Omega_\Lambda = 0.685$ (Planck 2018) - Standard Friedmann cosmology

No parameters are fitted to the BTFR data.

1.4 3. Test Against Observations

1.4.1 3.1 Data Sources

Survey	N	z range	Reference
KMOS3D	5	1.8–2.3	Übler et al. 2017

Survey	N	z range	Reference
SINS	3	2.0–2.2	Cresci et al. 2009
ALPINE	2	4.8–5.0	Faisst et al. 2020
Total	**10**	**1.8–5.0**	

1.4.2 3.2 Analysis Method

For each galaxy with measured (v_rot, M_bar, z):

1. **Local BTFR prediction:** log M_bar = 4 log v + 2.3
2. **Zimmerman prediction:** log M_bar = 4 log v + 2.3 – $\log_{10}E(z)$
3. **Residual:** Observed – Predicted
4. **χ^2:** Sum of (residual/error)^2

1.4.3 3.3 Results

```
Model                        χ²        χ²/dof
──────────────────────────────────────────────
Constant a₀ (local BTFR)    416.6     46.3
Evolving a₀ (Zimmerman)     118.7     13.2

Improvement: 3.5×
```

1.4.4 3.4 Interpretation

The constant a_0 model predicts high-z galaxies lie ON the local BTFR. They don't — they lie **below** it, exactly as the evolving a_0 model predicts.

1.5 4. Comparison to Other Models

1.5.1 4.1 ΛCDM Prediction

In ΛCDM, the BTFR arises from the baryon-halo connection. Evolution depends on: - Halo concentration evolution - Star formation efficiency - Gas fractions

No clear prediction for systematic BTFR offset with z.

1.5.2 4.2 Standard MOND

With constant a_0: - BTFR should be **identical** at all redshifts - **Falsified** by the observed offset

1.5.3 4.3 Evolving a_0

- Predicts specific offset: Δlog M = $-\log_{10}E(z)$

- **Matches observations** (3.5× better χ^2)

1.6 5. Falsification Criteria

This prediction is **falsified** if:

1.6.1 5.1 No Offset Observed

If JWST/ELT find:

Then there is no evolution, and the framework is wrong.

1.6.2 5.2 Wrong Sign

If high-z galaxies are **more** massive at fixed v (positive offset), the prediction fails completely.

1.6.3 5.3 Wrong Magnitude

If the offset is:

Then the specific form $a_0 \propto E(z)$ is wrong.

1.7 6. Future Tests

1.7.1 6.1 JWST NIRSpec

- Hα kinematics at z = 1–3
- Spatially resolved rotation curves
- Direct BTFR measurement at high z

1.7.2 6.2 ALMA

- CO/[CII] rotation curves at z = 2–6
- Gas masses from molecular lines
- Independent baryonic mass estimates

1.7.3 6.3 ELT HARMONI (2028+)

- 0.01” resolution spectroscopy
- Resolved kinematics at z = 2–3

- Definitive BTFR evolution measurement

1.8 7. What This Paper Does NOT Claim

1. **We do not claim to derive a_0 from first principles.** The relationship $a_0 = c\sqrt{(G\varrho_c)}/2$ has an empirically-determined coefficient.

2. **We do not claim this proves MOND.** The BTFR evolution follows from $a_0 \propto \sqrt{\varrho_c}$ regardless of the underlying theory.

3. **We DO claim** that IF a_0 tracks cosmic density, THEN the BTFR must evolve as predicted, and this can be tested.

1.9 8. Summary

1.9.1 The Prediction

1.9.2 The Test

Model	χ^2 (N=10)
Constant a_0	416.6
Evolving a_0	118.7
Improvement	**3.5×**

1.9.3 The Significance

- Specific, quantitative prediction
- Not fitted to the data
- Falsifiable with future observations
- Already favored by existing data

1.10 References

1. McGaugh, S.S., et al. (2000). "The Baryonic Tully-Fisher Relation." *ApJL*, 533, L99.

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4. Cresci, G., et al. (2009). "SINS survey." *ApJ*, 697, 115.

5. Faisst, A.L., et al. (2020). "ALPINE survey." *ApJS*, 247, 61.

6. Planck Collaboration (2020). *A&A*, 641, A6.

1.11 Reproducibility

Code: https://github.com/carlzimmerman/zimmerman-formula/tree/main/paper/analysis

```
import numpy as np

def E(z):
    return np.sqrt(0.315 * (1 + z)**3 + 0.685)

def btfr_offset(z):
    """BTFR mass offset at redshift z (in dex)"""
    return -np.log10(E(z))

# At z=2: offset = -0.48 dex (3× less mass at fixed v)
```

DOI: [To be assigned]

Code: https://github.com/carlzimmerman/zimmerman-formula